

```
In [32]: import pandas as pd
```

```
In [33]: pd.__version__
```

```
Out[33]: '2.3.3'
```

```
In [34]: emp = pd.read_excel(r'/Users/abhiramireddygali/Downloads/march(2025
```

```
In [35]: emp
```

```
Out[35]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [36]: id(emp)
```

```
Out[36]: 5977622800
```

```
In [37]: emp.head()
```

```
Out[37]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

```
In [38]: emp.tail()
```

```
Out[38]:
```

	Name	Domain	Age	Location	Salary	Exp
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

```
In [39]: emp.isnull().sum()
```

```
Out[39]: Name      0
        Domain    0
        Age       2
        Location   2
        Salary     0
        Exp       1
        dtype: int64
```

```
In [40]: emp.describe()
```

```
Out[40]:
```

	Name	Domain	Age	Location	Salary	Exp
count	6	6	4	4	6	5
unique	6	6	4	4	6	5
top	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
freq	1	1	1	1	1	1

```
In [41]: emp.columns
```

```
Out[41]: Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

```
In [42]: emp.shape
```

```
Out[42]: (6, 6)
```

```
In [43]: emp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         4 non-null     object
3   Location    4 non-null     object
4   Salary      6 non-null     object
5   Exp         5 non-null     object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
In [44]: emp.isnull()
```

```
Out[44]:
```

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

```
In [45]: emp
```

```
Out[45]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

Data Cleaning or Data cleansing

```
In [46]: emp['Name']
```

```
Out[46]: 0    Mike
1    Teddy^
2    Uma#r
3    Jane
4    Uttam*
5    Kim
Name: Name, dtype: object
```

```
In [47]: emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True)
```

```
In [48]: emp['Name']
```

```
Out[48]: 0    Mike
1    Teddy
2    Umar
3    Jane
4    Uttam
5    Kim
Name: Name, dtype: object
```

```
In [49]: emp['Domain'] = emp['Domain'].str.replace(r'\W', '', regex=True)
```

```
In [50]: emp['Domain']
```

```
Out[50]: 0    Datascience
         1      Testing
         2    Dataanalyst
         3      Analytics
         4    Statistics
         5          NLP
         Name: Domain, dtype: object
```

```
In [51]: emp['Age'] = emp['Age'].str.replace(r'\W','',regex =True)
```

```
In [52]: emp['Age']
```

```
Out[52]: 0    34years
         1    45yr
         2     NaN
         3     NaN
         4    67yr
         5    55yr
         Name: Age, dtype: object
```

```
In [53]: emp['Age'] = emp['Age'].str.extract('(\d+)')
```

```
In [54]: emp['Age']
```

```
Out[54]: 0    34
         1    45
         2    NaN
         3    NaN
         4    67
         5    55
         Name: Age, dtype: object
```

```
In [55]: emp
```

```
Out[55]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5^00#0	2+
1	Teddy	Testing	45	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67	NaN	30000-	5+ year
5	Kim	NLP	55	Delhi	6000^\$0	10+

```
In [56]: emp['Location'] = emp['Location'].str.replace(r'\W','',regex =True)
```

```
In [57]: emp['Location']
```

```
Out[57]: 0      Mumbai
         1      Bangalore
         2         NaN
         3      Hyderabad
         4         NaN
         5         Delhi
         Name: Location, dtype: object
```

```
In [58]: emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex = True)
```

```
In [59]: emp['Salary']
```

```
Out[59]: 0      5000
         1     10000
         2     15000
         3     20000
         4     30000
         5     60000
         Name: Salary, dtype: object
```

```
In [60]: emp
```

```
Out[60]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2+
1	Teddy	Testing	45	Bangalore	10000	<3
2	Umar	Dataanalyst	NaN	NaN	15000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5+ year
5	Kim	NLP	55	Delhi	60000	10+

```
In [61]: emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

```
In [62]: emp["Exp"]
```

```
Out[62]: 0      2
         1      3
         2      4
         3     NaN
         4      5
         5     10
         Name: Exp, dtype: object
```

```
In [63]: emp
```

Out[63]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [64]: `emp.head()`

Out[64]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5

In [65]: `clean_data = emp.copy()`

In [66]: `clean_data`

Out[66]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

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In [67]: `clean_data`

```
Out[67]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [68]: clean_data.isnull().sum()
```

```
Out[68]: Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

```
In [69]: clean_data['Age']
```

```
Out[69]: 0      34
1      45
2      NaN
3      NaN
4      67
5      55
Name: Age, dtype: object
```

```
In [70]: import numpy as np
```

```
In [71]: clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(c
```

```
In [72]: clean_data['Age']
```

```
Out[72]: 0      34
1      45
2     50.25
3     50.25
4      67
5      55
Name: Age, dtype: object
```

```
In [73]: clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(c
```

```
In [74]: clean_data['Exp']
```

```
Out[74]: 0      2
         1      3
         2      4
         3     4.8
         4      5
         5     10
         Name: Exp, dtype: object
```

```
In [75]: clean_data['Location'].isnull().sum()
```

```
Out[75]: np.int64(2)
```

```
In [76]: clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

```
In [77]: clean_data['Location']
```

```
Out[77]: 0      Mumbai
         1    Bangalore
         2    Bangalore
         3    Hyderabad
         4    Bangalore
         5      Delhi
         Name: Location, dtype: object
```

```
In [78]: clean_data
```

```
Out[78]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderabad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [79]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null      object
1   Domain       6 non-null      object
2   Age         6 non-null      object
3   Location    6 non-null      object
4   Salary      6 non-null      object
5   Exp         6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```



```
In [80]: clean_data['Age'] = clean_data['Age'].astype(int)
```

```
In [81]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int64
3   Location    6 non-null     object
4   Salary      6 non-null     object
5   Exp         6 non-null     object
dtypes: int64(1), object(5)
memory usage: 420.0+ bytes
```

```
In [82]: clean_data['Salary'] = clean_data['Salary'].astype(int)
```

```
In [83]: clean_data['Exp'] = clean_data['Exp'].astype(int)
```

```
In [84]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        6 non-null     object
1   Domain      6 non-null     object
2   Age         6 non-null     int64
3   Location    6 non-null     object
4   Salary      6 non-null     int64
5   Exp         6 non-null     int64
dtypes: int64(3), object(3)
memory usage: 420.0+ bytes
```

```
In [87]: clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Domain'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
```

```
In [88]: clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0    Name        6 non-null     category
1    Domain      6 non-null     category
2    Age         6 non-null     int64
3    Location    6 non-null     category
4    Salary      6 non-null     int64
5    Exp         6 non-null     int64
dtypes: category(3), int64(3)
memory usage: 938.0 bytes
```

```
In [89]: clean_data
```

```
Out[89]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [90]: clean_data.to_csv('clean_data.csv')
```

```
In [91]: import os
os.getcwd()
```

```
Out[91]: '/Users/abhiramireddygali'
```

```
In [92]: clean_data
```

```
Out[92]:
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [93]: import matplotlib.pyplot as plt
import seaborn as sns
```

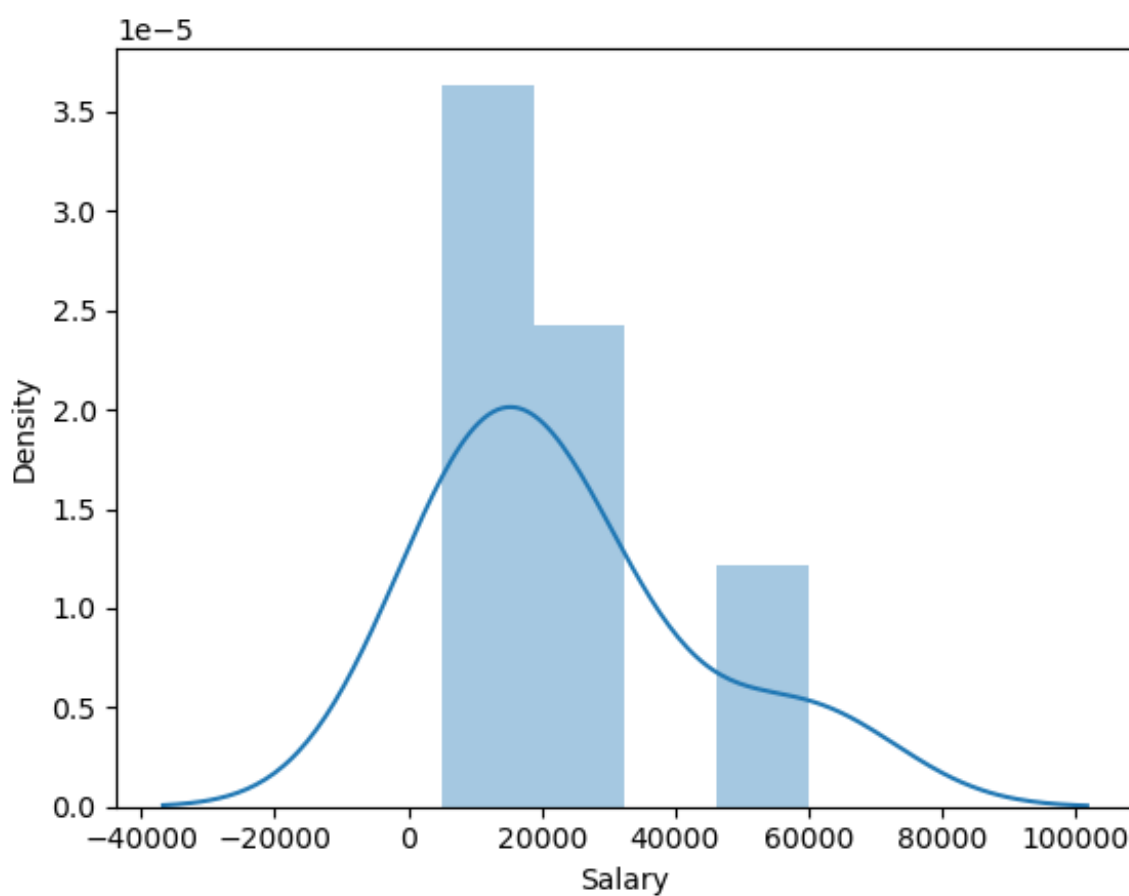
```
In [94]: import warnings
```

```
warnings.filterwarnings('ignore')
```

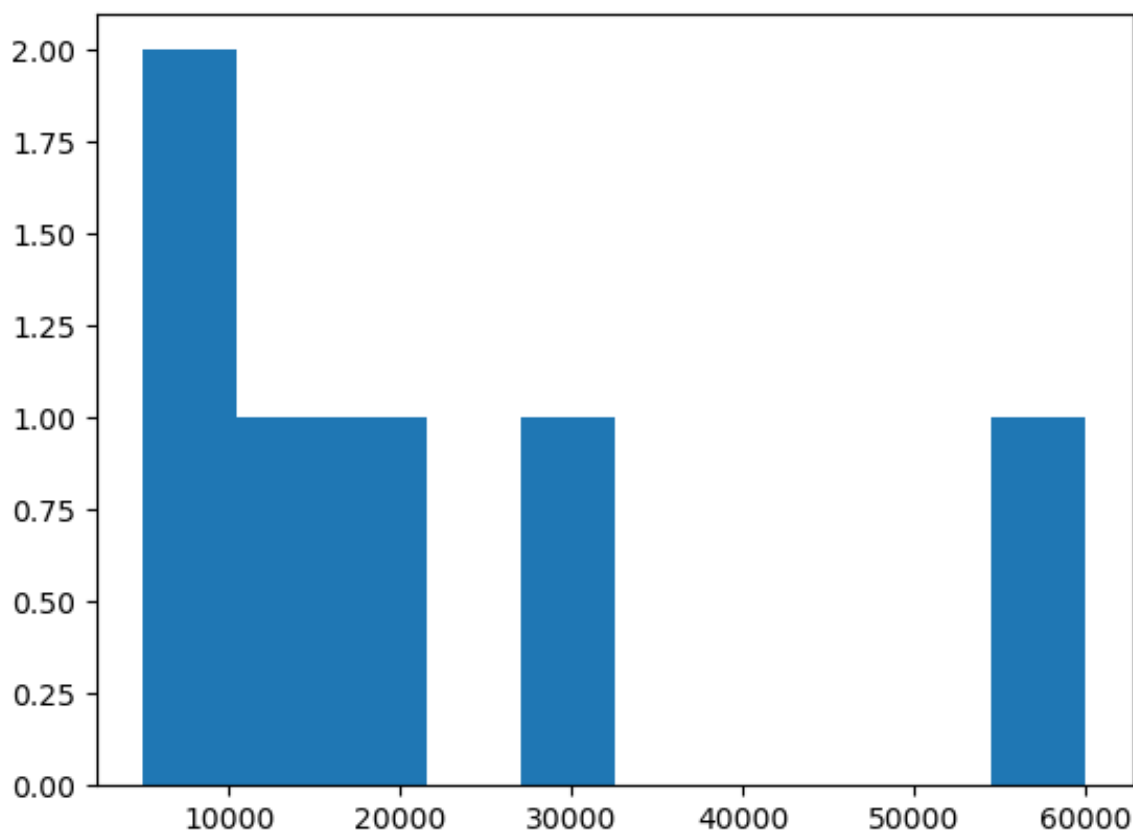
```
In [95]: clean_data['Salary']
```

```
Out[95]: 0    5000  
1   10000  
2   15000  
3   20000  
4   30000  
5   60000  
Name: Salary, dtype: int64
```

```
In [97]: vis1 = sns.distplot(clean_data['Salary'])
```



```
In [100... vis2 = plt.hist(clean_data['Salary'])
```

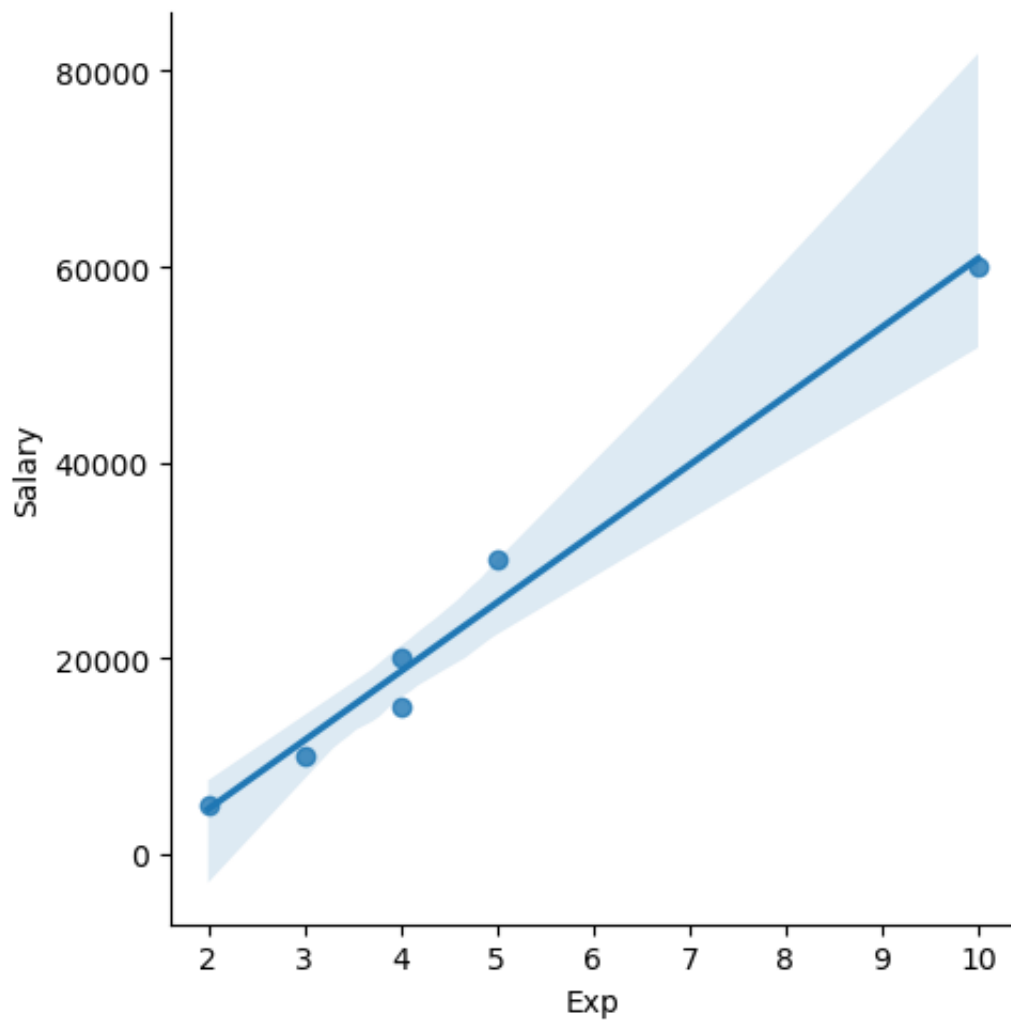


```
In [101... clean_data
```

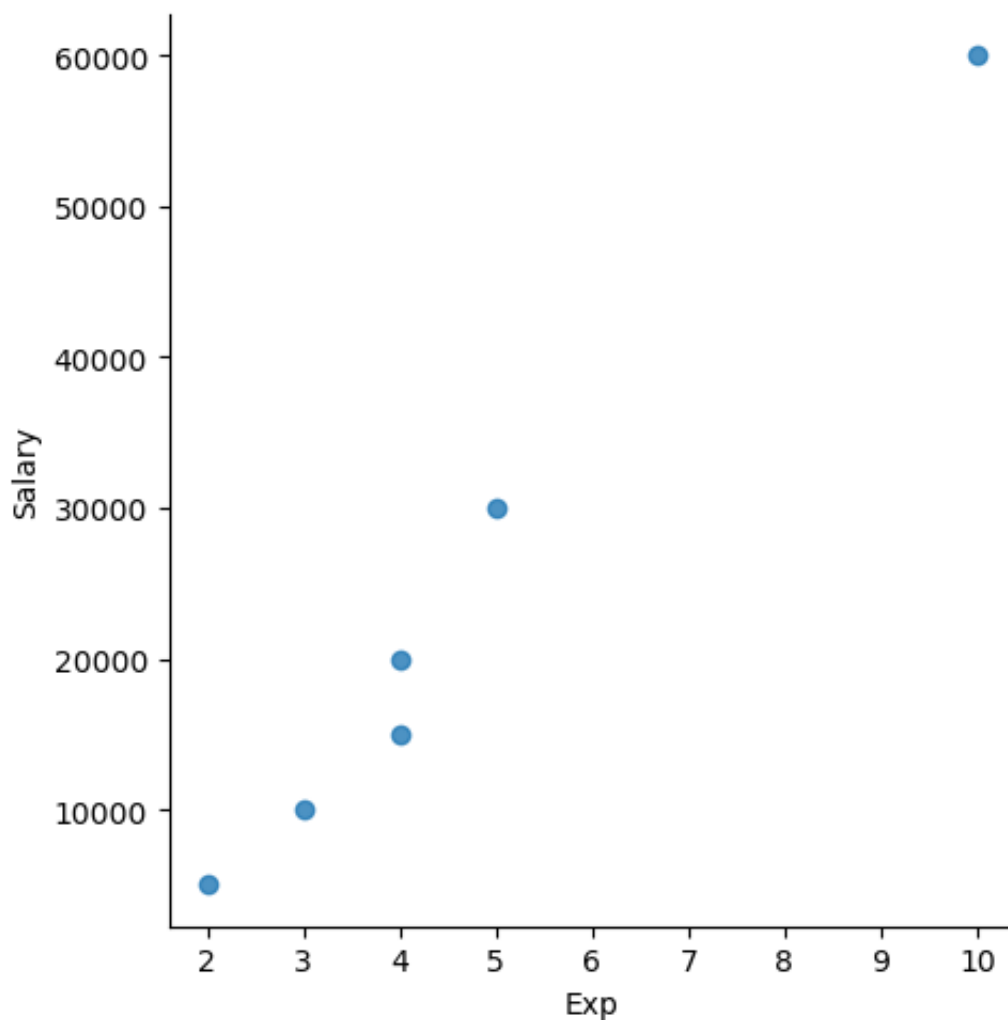
```
Out[101... 
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50	Bangalore	15000	4
3	Jane	Analytics	50	Hyderbad	20000	4
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

```
In [102... vis4 = sns.lmplot(data = clean_data,x = 'Exp',y = 'Salary')
```



```
In [103... vis4 = sns.lmplot(data = clean_data,x = 'Exp',y = 'Salary',fit_reg=
```



In [104... `clean_data[:: -1]`

Out[104...

	Name	Domain	Age	Location	Salary	Exp
5	Kim	NLP	55	Delhi	60000	10
4	Uttam	Statistics	67	Bangalore	30000	5
3	Jane	Analytics	50	Hyderabad	20000	4
2	Umar	Dataanalyst	50	Bangalore	15000	4
1	Teddy	Testing	45	Bangalore	10000	3
0	Mike	Datascience	34	Mumbai	5000	2

In [105... `clean_data[0:5:2]`

Out[105...

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
2	Umar	Dataanalyst	50	Bangalore	15000	4
4	Uttam	Statistics	67	Bangalore	30000	5

In [106... `x_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]`

In [107... x_iv

Out[107...

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50	Bangalore	4
3	Jane	Analytics	50	Hyderbad	4
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [110... y_dv = clean_data[['Salary']]

In [111... y_dv

Out[111...

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [112... imputation = pd.get_dummies(clean_data, dtype=int)

In [113... imputation

Out[113...

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Na
0	34	5000	2	0	0	1	0	
1	45	10000	3	0	0	0	1	
2	50	15000	4	0	0	0	0	
3	50	20000	4	1	0	0	0	
4	67	30000	5	0	0	0	0	
5	55	60000	10	0	1	0	0	

In [114... imputation.columns